

Curriculum Vitae

Kai Guo(郭凯)

PERSONAL INFORMATION

Family Name: Guo

Given Name: Kai

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CURRENT POSITION

University of North Dakota, School of Medicine and Health Sciences.

Nov.2016-Now Post-doctoral Research Fellow

EDUCATION

Huazhong Agricultural University (HZAU), Sep. 2008-Nov. 2015

M.S.E. and Ph.D in Biochemistry and Molecular Biology.

Huazhong Agricultural University (HZAU), Sep.2005-Jun.2008

Bachelor in Biotechnology Specialty

SCHOLARSHIPS, HONORS, AND AWARDS

Major award of The Dupont Scholarship, 2014

Excellent individual scholarship of BBRC, 2010,2011,2012

Excellent individual scholarship of College of Life Science and Technology,
2008,2009

Single-curriculum scholarship of College of Life Science and Technology,
2006,2007

COMPUTER SKILLS

Programming language: C++, Perl, R, Python, Shell, Php, SQL

Operating system: MS Windows, Unix, Linux, MacOS

Bioinformatics applications: Samtools, Picard, GATK, Bioconductor, HISAT, Stringtie, Tophat2, Cufflink, SOAPdenovo, BWA, Bowtie, BLAST, MEME, MACS, BLAT, FASTX, PacBio related tools and other Bioinformatics softwares

RESEARCH INTERESTS

1. Bioinformatics
2. Epigenetics
3. Transcriptomic analysis
4. Genome Assembly
5. Molecular evolution

PUBLICATIONS

1. Yang Wang, Xukai Li, **Kai Guo**, Liangcai Peng and Yanting Wang*. Gene Profiling of Plant Cell Wall Biosynthesis for Genetic Enhancing Biomass Enzymatic Saccharification in Cereal Crops. *J Plant Biochem Physiol*, 2017, 5: 179. doi: 10.4173/2329-9029.1000179
2. Xukai Li, **Kai Guo**, Xiaobo Zhu, Peng Chen, Ying Li, Guosheng Xie, Lingqiang Wang, Yanting Wang, Staffan Persson*, Liangcai Peng*. Domestication of rice has reduced the occurrence of transposable elements within gene coding regions. *BMC Genomics*, 2017, 18: 55. DOI: 10.1186/s12864-016-3454-z.
3. Ao Li, Ruyi Wang, Xianliang Li, Mingyong Liu, Jian Fan, **Kai Guo**, Bing Luo, Tingting Chen, Shengqiu Feng, Yanting Wang, Bingrui Wang, Liangcai Peng, Tao Xia*. Proteomic profiling of cellulase-aid-extracted membrane proteins for functional identification of cellulose synthase

complexes and their potential associated-components in cotton fibers. *Scientific Reports*, 2016.

4. Mingliang Zhang, Feng Wei, **Kai Guo**, Zhen Hu, Yuyang Li, Guosheng Xie, Yanting Wang, Xiwen Cai, Liangcai Peng, Lingqiang Wang*. A novel FC116/BC10 mutant distinctively alters the expression of the genes for wall polymer synthesis in rice. *Frontiers in Plant Science*, 2016.
5. Haiyan Sun, **Kai Guo**, Shengqiu Feng, Weihua Zou, Ying Li, Chunfen Fan, Liangcai Peng*. Positive selection drives adaptive diversification of the 4-coumarate: CoA ligase (4CL) gene in angiosperms. *Ecology and Evolution*, 2015, DOI: 10.1002/ece3.1613.
6. Fengcheng Li[§], Mingliang Zhang[§], **Kai Guo**, Zhen Hu, Ran Zhang, Yongqing Feng, Xiaoyan Yi, Weihua Zou, Lingqiang Wang, Changyin Wu, Jinshan Tian, Tigang Lu, Guosheng Xie*, Liangcai Peng*. High-level hemicellulosic arabinose predominately affects lignocellulose crystallinity for genetically enhancing both plant lodging resistance and biomass enzymatic digestibility in rice mutants. *Plant Biotechnology Journal*, 2015, 13:514-525
7. **Kai Guo**, Weihua Zou, Yongqing Feng, Mingliang Zhang, Jing Zhang, Fen Tu, Guosheng Xie, Lingqiang Wang, Yangting Wang, Sebastian Klie, Staffan Persson Liangcai Peng*. An integrated genomic and metabolomic framework for cell wall biology in rice. *BMC Genomics*, 2014, 15: 596.
8. Xianliang Li, Tao Xia*, Jiangfeng Huang, **Kai Guo**, Xu Liu, Tingting Chen, Wen Xu, Xuezhe Wang, Shengqiu Feng, Liangcai Peng*. Distinct biochemical activities and heat shock responses of two UDP-glucose sterol glucosyltransferases in cotton. *Plant Science*, 2014, 219-220: 1-8.
9. Zhiliang Wu[§], Mingliang Zhang[§], Lingqiang Wang*, Yuanyuan Tu, Jing Zhang, Guosheng Xie, Weihua Zou, Fengcheng Li, **Kai Guo**, Qing Li, Chunbao Gao, Liangcai Peng*. Biomass digestibility is predominantly affected by three factors of wall polymer features distinctive in wheat accessions and rice mutants. *Biotechnology for Biofuels*, 2013, 6:183.
10. Haiyan Sun, Ying Li, Shengqiu Feng, Weihua Zou, **Kai Guo**, Chunfen Fan, Shengli Si, Liangcai Peng*. Analysis of five rice 4-coumarate:coenzyme a ligase enzyme activity and stress response for potential roles in lignin and flavonoid biosynthesis in rice.

Biochemical and Biophysical Research Communications, 2013, 430(3):1151-1156.

11. Guosheng Xie[§], Bo Yang[§], Zhengdan Xu, Fengcheng Li, **Kai Guo**, Mingliang Zhang, Lingqiang Wang, Weihua Zou, Yanting Wang, Liangcai Peng*. Global identification of multiple OsGH9 family members and their involvement in cellulose crystallinity modification in rice. *PLoS One*, 2013, 8(1)
12. Lingqiang Wang[§], **Kai Guo**[§], Yu Li, Yuanyuan Tu, Huizhen Hu, Bingrui Wang, Xiaocan Cui and Liangcai Peng*. Expression profiling and integrative analysis of the CESA/CSL superfamily in rice. *BMC Plant Biology*, 2010,10:282-298.

EXTRACURRICULAR ACTIVITIES

Sep.2014: Organizer of HZAU-UCD Joint Symposium: From Plant Growth to Biomass Products

Nov.2013: Organizer of Sion-AUS Symposium on Biotechnology and Bioenergy

Nov.2012: Organizer of the third international symposium on Bioenergy and Biotechnology

Sep.2013—now: Organizer and Teacher of Bioinformatics Learning group of HZAU

WORK EXPERIENCES

Mar. 2013 — Mar. 2014: Assistant Bioinformatics Engineer of Katie Wuhan Engineering Technology Research Institute Co.